

Stool Collection, Processing and Storage

9.1 Overview

Ghana Breast Health Study participants will be asked to provide a stool sample. The stool collection for cases began at KBTH on December 11, 2013, at KATH on November 5, 2013, and at PLH on November 14, 2013 as a pilot study to test the collection procedures. Stool samples from Controls began with the second round of Control accrual (the selected list of Controls from the second enumeration) in June 2014. Stool samples are collected only from participants who are willing and able to provide them.

From the start of the implementation of the stool protocol to February 2015, stool samples were collected only in the hospital at the time of the study visit. Beginning in February 2015, home collection can be attempted from Controls who consent to give a stool sample but are not able to provide one at their study visit at the hospital. The protocol for home collection is outlined in Section 9.3.2. Staff may re-contact Controls from the second enumeration list who were willing but unable to provide stool samples at their study visit, and attempt to collect stool samples at home. Controls from the pilot phase (first enumeration) will not be re-contacted for stool collection.

The Biospecimen Master Kit contains a Stool Collection Kit with a yellow circle sticker. This kit contains two pre-labeled, brown-capped vials that will be used to store the stool specimen. The Master Kit also contains a yellow Stool Collection Form. Study staff will complete a yellow Stool Collection Form for each participant who agrees to any biospecimen collection and who is asked to give a stool sample, regardless of whether a stool sample was collected.

After stool samples have been collected, samples will be sent to the laboratory where one vial will be snap frozen at -80°C immediately upon receipt, and one vial will have RNALater added and then be frozen at -80°C . Laboratory staff will add RNALater upon receipt at the lab per the processing protocol (Section 9.4); other Study Staff will not handle the RNALater. Both samples will remain stored at -80°C until shipped to the United States.

9.2 Completing the Stool Collection Form

The Participant ID (PID) label should already have been placed on the yellow Stool Collection Form (Appendix 9-1) by the Study Manager or Interviewer in preparation for collecting biospecimens. If the PID label is missing, place it on the form in the upper left hand corner. Verify that the root Sample ID (GBR#####) on the Stool Collection Form (GBR#####-SX01) matches the Sample ID on both Stool Collection Tubes (GBR#####-2011, -2012).

The Interviewer will confirm that the participant has provided consent to participate in the study and is willing to provide a stool sample. If the participant did not initial on the Informed Consent that she agrees to provide a stool sample, then do not collect the sample. Regardless of whether or not a stool sample is collected, the study staff member should complete the appropriate questions contained in the shaded Collection Summary Box per the instructions below and in Appendix 9-2

The yellow Stool Collection Form has three separate sections: the gray shaded Collection Summary Box with Questions A-H, Section 1 with Questions 1-4 and the gray shaded Processing Summary Box with Questions I and J. Study staff will complete these sections at different times during the stool collection process.

If a participant does not consent to stool collection, study staff will complete only questions A and B in the shaded Collection Summary Box. All other questions will be left blank. If the participant does agree to provide a stool sample, study staff will complete items A-H in the gray shaded Collection Summary Box, ask the participant questions 1-4 and record responses. The laboratory staff will complete Item J.

Specific instructions on stool form completion:

- Items A and B complete for all participants approached for stool collection, even if they refuse stool collection.
- Items C (Kit Distributed to Participant) and D (Kit Receipted or Closed Out) may be the same date, particularly if the collection is at the hospital. If the participant is a Case who has been admitted to the hospital, then it is possible that the stool collection kit will be

provided to the participant on one day and receipted by study staff on another. Home collections will also have different dates for Items C and D.

- Item G in the Collection Summary Box will be answered “Home.....1” for stool samples collected at home and “Hospital.....2” for stool samples collected at the hospital.
- Item H is applicable for home visits only. This item should be left blank if the specimen was collected in the hospital. For stool samples collected at home, Item H should be completed as the date and time the stool samples were placed in the cooler for transport to the laboratory.
- Study staff will ask participants Questions 1-4 in Section 1 when the stool sample is received. Questions 1 through 4 pertain to the participant’s use of antibiotics and her menstrual cycle.
- Item I is not applicable - the tubes will not be placed in a 4° C refrigerator. This item should be left blank.
- Laboratory staff will complete Item J in the gray shaded Processing Summary Box when the stool samples are placed in the freezer.

9.3 Stool Collection Procedures

The Stool Collection kit contains two labeled, brown-capped collection vials. Study staff will verify that the specimen IDs on the vials match the base sample ID on the Master Kit before giving them to the participant. The vials will be empty; no RNALater will be added to the vials until the sample arrives at the laboratory.

9.3.1 Hospital Collection

Collection of the stool sample in the hospital during the study visit is the preferred mode of collection and should be attempted first for all Cases and Controls who agree to stool collection.

Study staff at each site will need to develop procedures appropriate to their institution to distribute kits, follow-up with participants or provide participants with a way to alert study staff, and retrieve specimens from participants who have been admitted to the hospital.

When scheduling study visits for Controls, study staff should inform controls that they will be asked to provide a stool sample during their visit and to avoid passing stool earlier that day if possible. Study staff will give the vials to the participants at the hospital and ask participants to attempt to collect in the hospital. Study staff should explain how to collect the stool sample using demonstration items and the procedures provided below.

Instructions For Participants To Collect the Stool Sample

- Do not pass the stool directly into the toilet.
- Avoid contaminating your stool sample with your urine; if possible urinate before placing the collecting device in place to catch the stool.
- Do not let urine or toilet bowl water mix with the stool when passing the stool or collecting it.
- Open each Stool Collection Tube carefully.
- Use the scoopula in each vial to transfer a single scoop of the stool sample from the collection into each of the two vials.



- Close the tube and make sure it is secure.
- Wash hands thoroughly.
- Return the specimen vials to the study staff member.

Study staff should ask the participant if she has any questions regarding the instructions for collecting the stool sample.

After the participant returns the stool sample to the study staff member, the staff member will complete items D through H in the gray shaded Collection Summary Box, and ask the participant Questions 1-4 in Section 1.

Study staff will deliver the vials to the laboratory directly as soon after collection as possible.

9.3.2 Home Collection

If the participant is not able to provide a stool sample during the study visit at the hospital, study staff may give the stool collection kit to the participant to collect a specimen at home. The study staff member and the participant should coordinate the day on which the participant will collect the sample and the staff member will come to the participant's home to retrieve it. The sample should be provided on the same day that it is retrieved to minimize time between collection and processing/storage.

Below is a suggested protocol for home collection. Each site will need to determine the logistics that work best for their circumstances.

- 1) Provide the participant with the stool collection kit labeled with their sample ID.
- 2) Record Items A-C on the Stool Collection Form and retain the form (do not give the form to the participant).
- 3) Study staff and participant will agree on a day for stool collection.
- 4) Participant will collect stool sample on the morning of the agreed upon day.
- 5) Retrieve the sample on the agreed upon day of collection, and bring the Stool Collection Form to the retrieval.
- 6) Ensure that the tops of the vials are secure.
- 7) Place the vials in a Ziploc plastic bag in a cooler with ice (not dry ice, just regular ice) and submerge in the ice to cool the sample as quickly as possible.
- 8) Ask the participant questions 1 through 4 on the Stool Collection Form and record responses.
- 9) Record items D- H on the Stool Collection Form.
- 10) Transport the vials to the laboratory on the same day with as little time lapse as possible.

If any problems are encountered with the stool collection or with completing the Stool Collection Form, bring them to the Study Manager's attention as soon as possible.

9.4 Stool Sample Processing and Storage

Processing and storage will be the same for stool samples collected in the hospital and at home.

9.4.1 Processing

The tubes will be processed as follows:

- GBR#####-2011 will have RNALater added by Study staff prior to being placed in a -80° C freezer. This is used to preserve the specimen.
- GBR#####-2012 will be placed in a – 80° C freezer upon receipt and will not have RNALater added.

Instructions for Study Staff to Fill Vial GBR#####-2011 with RNALater After Receiving Sample

- Study staff should wear gloves when working with or handling a vial containing the RNALater solution. This is a part of universal precautions to prevent exposure to the chemicals contained in the RNALater +antibiotic pre-mixed solution.
- Using one of the 5 ml pipettes provided, transfer 5ml of the RNALater + antibiotic pre-mixed solution from the 50ml falcon tube to the pre-labeled vial GBR#####-2011. Discard the pipette after use.
- Securely cap and shake the vial to mix the contents thoroughly.
- When not in use, the RNALater solution should be stored in a refrigerator at 4° C.

9.4.2 Storage

Place both collection vials in a 3-inch storage box with a 5x5 grid in a -80° C freezer for storage until shipped to the NCI Biorepository. The samples will not be stored in a 4° C refrigerator prior to freezing.

Record the date and time the stool samples were placed in the freezer on the Stool Collection Form (Processing Summary Box, Item J). Item I on the form is not relevant and should be left blank.

9.5 Form Review

Study staff should check the stool forms for completeness and legibility as follows:

- 1) If a stool sample was collected, all items on the form should be completed except Item I.
- 2) If the sample was collected at the hospital, Item H should be blank as well.
- 3) If the subject did not consent to stool sample collection, only items A and B should be completed.
- 4) If consent was given but no stool sample was collected, only items A through F should be completed.

The Study Manager will document the result of the stool collection in the Recruitment Tracking Log. In the 'Stool' column on the Recruitment Tracking Log record:

- 'Y' if a sample was obtained
- 'N' if the participant consented but a sample not obtained
- 'P' if the participant consented and sample collection is pending
- 'No Consent' if the participant refused to provide consent for the stool collection
- Leave blank if the participant was enrolled prior to the initiation of the stool protocol so NCI and Westat will know those subjects were not approached for stool collection.